The relative stability of the alkyl carbocations and carbanions are given below.

Relative stability carbocations.

$^{+}C(CH_3)_3 > {}^{+}CH(CH_3)_2 > {}^{+}CH_2CH_3 > {}^{+}CH_3$

relative stability of carbanions

$^{-}C(CH_3)_3 < {}^{-}CH(CH_3)_2 < {}^{-}CH_2CH_3 < {}^{-}CH_3$

The energy required to bring about homolytic splitting is greater than that of heterolytic splitting.

12.1.3 Nucleophiles and elctrophiles

Nucleophiles are reagents that has high affinity for electro positive centers. They possess an atom has an unshared pair of electrons, and hence it is in search for an electro positive centre where it can have an opportunity to share its electrons to form a covalent bond, and gets stabilised. They are usually negatively charged ions or electron rich neutral molecules (contains one or more lone pair of electrons). All Lewis bases act as nucleophiles.

Types	Examples	Electron rich site
Neutral molecules having unshared pair of electron	Ammonia (NH_3) and amines (RNH_2)	N:
	Water (H_2O), alcohols (ROH) and ethers (R-O-R)	:O:
	Hydrogen sulphide (H_2S) and thiols (RSH)	:S:
Negatively charged nucleophiles	Chlorides (Cl^-), bromides (Br^-) and iodides (I^-)	X^-
	Hydroxide (HO^-), alkoxide (RO^-) and Carboxlate ions ($RCOO^-$)	O^-
	Cyanide (CN^-)	N^-

Electrophiles are reagents that are attracted towards negative charge or electron rich center. They are either positively charged ions or electron deficient neutral molecules. All Lewis acids act as electrophiles. Neutral molecules like $SnCl_4$ can also act as an electrophile, as it has vacant d-orbitals which can accommodate the electrons from others.

Types	Examples	Electron deficient entity
Neutral electrophiles	Carbon dioxide (CO_2), dichlorocarbene ($:CCl_2$)	C
	Aluminium chloride ($AlCl_3$), boron trifluoride (BF_3) and ferric chloride ($FeCl_3$)	Metal (M)
Positively charged electrophiles	Carbocations (R^+)	C^+
	Proton (H^+)	H^+
	Alkyl halides (RX)	X^+
	Nitrosonium ion (NO^+)	O^+
	Nitronium ion ($^+NO_2$)	N^+

Unit 12.indd 164 23-12-2021 17:15:06